



CS 40700

Sprint1 Planning

Team 16

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Sprint Overview:

In Sprint 1, our primary focus is to establish the foundation of the game, ensuring that the core structure and fundamental mechanics are in place. This includes setting up both the frontend (Unity) and backend (Node.js) frameworks, implementing basic UI navigation, and ensuring smooth interactions between the client and server. By the end of the sprint, the game should have a functional main menu, loading screen, and basic user authentication system, allowing players to create accounts, log in, and manage save files.

One of the key components of this sprint is the development of card-based combat mechanics. This involves creating a turn-based system, defining card attributes, and implementing basic resource management features such as stamina consumption and regeneration. Players should be able to draw and play cards, with proper UI feedback to indicate changes in health, resources, and card piles.

The educational aspect of the game will also begin development in this sprint, focusing on teaching the printf function through interactive NPC dialogues and mini-games. Players will engage in guided tutorials and challenges that gradually increase in complexity, reinforcing their understanding of formatting outputs. To enhance engagement, the first achievement system will be introduced, rewarding players for mastering key concepts.

User experience is another major focus, with efforts dedicated to creating smooth transitions and polishing UI interactions. Features like a skippable loading animation, intuitive button navigation, and real-time feedback will ensure a seamless and engaging experience. Save file management will also be implemented, allowing users to create, load, and delete multiple save files for different game progressions.

Sprint Period:

- Feb. 3 - Feb. 27

Scrum Master:

- Yuchuan Huang

Scrum Meeting:

- Monday 11:30am - 1:30pm
- Thursday 7:00pm - 9:00pm

Risk and Challenges:

During Sprint 1, we encountered several challenges. One of the main issues was the integration between the front-end (Unity) and back-end (Node.js), which was crucial for ensuring smooth data transfer. This included tasks like user authentication, managing game archives, and synchronizing the game state. We had to carefully design APIs and optimize data flow to ensure everything worked seamlessly.

Another challenge came with developing the turn-based battle system. This involved managing cards, calculating resource consumption, and creating enemy AI logic. We had to make sure the system was smooth while maintaining a balanced gameplay experience, so players wouldn't feel frustrated by any imbalances in the battle mechanics.

Performance optimization was also a significant focus. For example, the loading animations needed to be immersive, but we had to ensure that players could skip them for a quicker transition to the main interface. Additionally, optimizing database queries was important to avoid any bottlenecks, especially for accessing archive and card data, which could slow down the game if not handled efficiently.

From a user experience perspective, integrating educational content like the "printf" teaching module was challenging. We needed to design it to be interactive and engaging, to avoid boring the players and keep them motivated to learn.

Nonetheless, none of us had prior experience with Unity, Node.js, or MongoDB. This lack of familiarity with the technologies meant that we had to spend additional time learning and troubleshooting as we progressed. Each team member had to quickly get up to speed with new concepts and frameworks, which slowed down development at times. We had to rely heavily on research, online tutorials, and trial and error to solve issues, especially when it came to handling the integration between the front-end, back-end, and database. This steep learning curve was a challenge, but it also helped us grow and gain a deeper understanding of the technologies we're working with.

Time management was a key consideration in Sprint 1. We had to prioritize core features like the UI, archive management, and battle system to make sure the essentials were ready first. Since team members were working on different modules simultaneously, maintaining smooth collaboration for code merging, testing, and version control was vital to avoid conflicts and integration issues.

Current Sprint Detail:

User Story #1

As a user, I want to see a loading animation when the game starts to make the experience feel engaging and polished, but I also want the option to skip the loading animation and directly access the main menu to save time when re-entering the game.

#	Description	Estimated Time	Owner
1	Front-end framework initialization and learning	1.5hr	Xiuwen Fnag
2	Backend framework initialization and learning	1.5hr	Xiuwen Fang
3	Prepare to load animation video	1hr	Xiuwen Fang
4	Importing videos into Unity and configuring the Video Player	2hr	Xiuwen Fang
5	Implement control of video playback and logic	2hr	Xiuwen Fang
6	Processing Skip UI and logic	2hr	Xiuwen Fang
7	Testing	1hr	Xiuwen Fang

Acceptance Criteria

1. Given the video is imported into Unity, when the developer configures the video player, then the video player must be able to play videos as expected.
2. Given the user opens the game, when the application starts, then the splash screen animation must be displayed first and play smoothly before transitioning to the main interface.
3. Given the splash screen animation is playing, when the animation starts, then the "Skip" button must be visible and accessible to the user throughout the animation.
4. Given the splash screen animation is playing, when the user clicks the "Skip" button, then the animation must stop immediately, and the user must be directed to the login page.

User Story #2

As a user, I would like to see a main page after loading the welcome animation, including the function buttons I need(settings, new age, load etc.).

#	Description	Estimated Time	Owner
1	UI of main page	2hr	Xiuwen Fang
2	Jump logic for all function buttons	1.5hr	Xiuwen Fang
3	Testing	0.5hr	Xiuwen Fang

Acceptance Criteria

1. Given the user has completed watching or skipping the welcome animation, when the main page loads, then all function buttons (e.g.settings, new age, load etc.) must be visible, styled correctly, and functional.
2. Given the main page is displayed, when the user clicks the "Load" button, then the application must navigate to the loading page.
3. Given the main page is displayed, when the user clicks the "New Game" button,the game must start from the beginning
4. Given the main page is displayed, when the user clicks the "Settings" button, then the application must navigate to the settings page.

User Story #3

As a user, I would like to create my own account.

#	Description	Estimated Time	Owner
1	UI design for account creation	2hrs	Xinyi Wu
2	Implement account buttons & logic	2hrs	Xinyi Wu
3	Database setup for user accounts	6hrs	Xinyi Wu
4	Connect frontend to backend	2hrs	Xinyi Wu
5	Backend updates for account logic	1hr	Xinyi Wu
6	Testing and debugging	2hrs	Xinyi Wu

Acceptance Criteria

1. Given the user is on the account creation page, when they enter valid information and submit, then the account must be created successfully.
2. Given the user tries to create an account with an already registered username, when they submit the form, then an error message must be displayed indicating that the username is taken.
3. Given the user created an account, when the account is successfully created, then they are directed back to the login page.

4. Given the user has successfully created an account, when they return later and access their accounts, their account information should be stored and displayed correctly.

User Story #4

As a user, I would like to log in and log out of my own account.

#	Description	Estimated Time	Owner
1	UI design for login/logout	2hrs	Xinyi Wu
2	Implement login/logout buttons & logic	2hrs	Xinyi Wu
3	Backend authentication logic	3hrs	Xinyi Wu
4	Connect frontend to backend	1hrs	Xinyi Wu
5	Database query for user authentication	3hrs	Xinyi Wu
6	Notification of login and logout	1hr	Xinyi Wu
7	Testing and debugging	1hr	Xinyi Wu

Acceptance Criteria

1. Given the user is on the login page, when they enter valid credentials and submit, then they can log into their account successfully.
2. Given the user enters incorrect credentials, when they try to log in, then an error message must be displayed indicating invalid credentials.
3. Given the user is logged in, when they click the "Logout" button, then they must be logged out and redirected to the login or main page.
4. Given the user is logged in, when they try to re-login into the same account, a notification of error login would pop out and the new login session is unsuccessful.

User Story #5

As a user, I want to earn my first achievement, "Person You Know Who," by demonstrating mastery of the printf function and shown on one achievement page, so I feel motivated to continue learning.

#	Description	Estimated Time	Owner
1	UI design for achievement system	2hrs	Xinyi Wu
2	Backend setup for storing achievements	1hr	Xinyi Wu

3	Connect frontend to backend	1hr	Xinyi Wu
4	Notification of getting the achievement	1hr	Xinyi Wu
5	Frontend display of locked achievement and unlocked achievement	1hr	Xinyi Wu
6	Testing and debugging	1hr	Xinyi Wu

Acceptance Criteria

1. Given the user successfully completes the tasks related to printf, when they demonstrate mastery, then a prompt will notify them that they have unlocked the “Person You Know Who” achievement.
2. Given the user is logged in, when they access a button of directing to the achievement page, then the achievement page will display all achievements.
3. Given the user has completed the achievement, when they access their achievement page, then the page will show the completed achievements along with their completion dates.
4. Given the user has not completed the achievement, when they check their achievement page, then the uncompleted achievements will display as unlock state.
5. Given the user completes an achievement, when they unlock the achievement, then a pop-up notification will appear to display the reward and notify them of the accomplishment.

User Story #6

As a user, I want each card to have a specific amount of cost, so I need to carefully manage my resources during each turn.

#	Description	Estimated Time	Owner
1	Add cost attribute to card class	1 hrs	Yuchuan Huang
2	Implement a method that checks if the player's current resource pool is $\geq$ the card, If insufficient resources, block the play action and give feedback (e.g., a “not enough resources” message).	3 hrs	Yuchuan Huang
3	Once the card is confirmed playable, subtract the card's cost from the player's resources.	2 hrs	Yuchuan Huang
4.	Testing	1 hrs	Yuchuan Huang

Acceptance Criteria

1. Given that a card has a cost attribute, when the card is displayed in the hand, then its cost is visible, so the player knows the resource requirement.
2. Given that the player attempts to play a card, when the player's current resources are less than the card's cost, then the game prevents the card from being played and displays a warning.
3. Given that the player's resources are greater than or equal to the card's cost, when the player plays the card, then the card's cost is deducted from the player's resource pool, and the card effect triggers.

User Story #7

As a player, I want to see my health, enemy health, and card details during combat so that I can make informed decisions.

#	Description	Estimated Time	Owner
1	Create UI for player HP, enemy HP, and a hand area to display cards.	4hrs	Yuchuan Huang
2	For each card in hand, show a small UI panel with the card's name, cost, and effect.	4hrs	Yuchuan Huang
3	When a card is played, show that it is removed from hand	3 hrs	Yuchuan Huang
4	Update the HP bars in real time after damage is dealt.	2 hrs	Yuchuan Huang
5	Create health mechanism for enemy and player	2 hrs	Yuchuan Huang
6	End combat if player HP ≤ 0 (defeat) or enemy HP ≤ 0 (victory).	2 hrs	Yuchuan Huang
7	Update the combat result in database	1hr	Xinyi Wu
8.	Testing	1hr	Yuchuan Huang

Acceptance Criteria

1. Given that combat is active, when the combat UI is displayed, then the player's HP and the enemy's HP are both visible on the screen (via health bars or text).

2. Given that the player has cards in hand, when those cards are displayed on screen, then each card shows its name, cost, and primary effect (or effect type).
3. Given that a card is played, When the effect is triggered, then the card is visually removed from the hand area.
4. Given that damage is dealt to the player or the enemy, when health values change, then the health bars/indicators update immediately to reflect the new HP totals.
5. Given that the player or the enemy takes damage, when that entity's health is reduced to 0 or below, then the combat ends and shows the end screen.
6. Given that an entity's health is reduced to 0 or below, when the game ends, then no further action (e.g., playing cards or attacking) can be performed.

User Story #8

As a user, I want a basic deck of cards so that I can draw and play cards in combat.

#	Description	Estimated Time	Owner
1	Create a class for cards with properties such as name, cost, effectType, etc.	2 hrs	Yuchuan Huang
2	add a basic manager for card information management	2 hrs	Yuchuan Huang
3	Implement a function to draw cards from "draw pile" into "hand"	2 hrs	Yuchuan Huang
4	Implement a function to move card from "hand" into "discard pile" in the proper time	1 hr	Yuchuan Huang
5	Implement a function to move all the card from "discard pile" into "draw pile" when "draw pile" don't have any cards	1 hr	Yuchuan Huang
6	Implement a function to shuffle the draw pile	1 hr	Yuchuan Huang
7	Implement the function to remove the card from the hand when played	1 hr	Yuchuan Huang
8	Implement the function that Triggers the card's effect (e.g., dealing damage, adding a shield, etc.).	2 hrs	Yuchuan Huang
9	Save the card information in database	2hrs	Xinyi Wu

10	Testing	1 hr	Yuchuan Huang
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Acceptance Criteria

1. Given that the player starts a new combat, when the combat begins, then a deck of cards is loaded for the player
2. Given that the player has a draw pile, when the player draws cards at the start of their turn, then the specified number of cards is moved from the draw pile to the player's hand.
3. Given that cards in hand are played or discarded, when a card is removed from the hand, then it is placed into the discard pile.
4. Given that the draw pile is empty, when a new card draw is required, then all cards from the discard pile are moved to the draw pile, and the draw pile is shuffled
5. Given that a card is "played," when the card's effect is triggered, then the appropriate outcome occurs (e.g., dealing damage, adding shield).

User Story #9

As a player, I want to engage in a simple turn-based combat system so I have time to plan my strategy and respond to the boss's moves.

#	Description	Estimated Time	Owner
1	Create a simple turn manager that tracks whose turn it is.	2 hrs	Yuchuan Huang
2	Trigger enemy action during the enemy's turn.	2 hrs	Yuchuan Huang
3	Add an "end" button that ends the player's current turn when the player clicks it.	2 hrs	Yuchuan Huang
4	Testing	1 hr	Yuchuan Huang

Acceptance Criteria

1. Given that the combat has started, when the turn manager initializes, then it sets the turn order to begin with whoever is faster
2. Given that the combat has started, when the combat is in progress, then the turn manager will track who the current turn belongs to.
3. Given it is the enemy's turn, when the turn manager signals the enemy to act, then the enemy performs its action (e.g., dealing damage) and then ends its turn.

4. Given that combat is active, when it is the player's turn, the player can take action (e.g., play cards).
5. Given it is the player's turn, when the player clicks the "End Turn" button, then the turn manager switches to the enemy's turn and prevents further player actions until the next player turn.

User Story #10

As a user, I want the game to have NPC interactions to teach me the basics of the printf function, so I can use them effectively.

#	Description	Estimated Time	Owner
1	Adding the NPC and user's figure	1hr	Xinyi Guan
2	Writing the dialogue script for NPC	1hr	Xinyi Guan
3	Adding the interaction logic with NPC(enter for next and dialogue log)	4hrs	Xinyi Guan
4	Pause game buttons and later choices(e.g. continue current game or back to home page)	3hr	Xinyi Guan
5	Adding a background for NPC interaction	1hr	Xinyi Guan
6	Design and find NPC logo/figure	2hr	Xinyi Guan
7	Design and find background figure	2hr	Xinyi Guan
8	Testing	1hr	Xinyi Guan
9	Learning and Setting up all framework environments.	1hr	Xinyi Guan

Acceptance Criteria

1. Given the user starts a new game, when the user initialized successfully, then the printf teaching dialog will appear.
2. Given the NPC has a dialogue script, when the user presses "Enter," then the next dialogue line should appear.
3. Given the game has pause buttons, when the user clicks this button, then the game must pause and choose whether to "Continue", "Back to Main Page" .
4. Given the user clicks the "Continue" button, then the dialog will keep going on.
5. Given the user clicks "Back to Main Page", then the user will be transferred to the Main page.
6. Given a background is designed for interactions, when the user enters an interaction, then the interaction screen must display the background distinctively.

7. Given the user wants to review past conversations, when the user clicks the “review dialog”, then the full dialogue log must be accessible and be opened.

User Story #11

As a user, I would like mini-games to practice printf with increasingly complex printf challenges, so I can gradually improve my skills.

#	Description	Estimated Time	Owner
1	Adding a keyboard-input box to let player input the command	2hrs	Xinyi Guan
2	Adding the logic for printf(): if syntax correct, print accordingly; else displays garbled characters.	4hrs	Xinyi Guan
3	Teaching module: printf with string/int/double(TBD)	3hr	Xinyi Guan
4	Testing module: player input and practicing specific styles.	2hr	Xinyi Guan
5	Task for player self-naming using the correct printf() syntax.	1hr	Xinyi Guan
6	Connect frontend to backend	1hr	Xinyi Guan
7	Update the user defined name in database	1hr	Xinyi Wu
8	Update the user game process in database(finished or not finished)	1hr	Xinyi Wu
9	Testing	1hr	Xinyi Guan

Acceptance Criteria

1. Given a keyboard input box is implemented, when the player enters a correct or incorrect printf command, then the system must either print the output or display garbled characters.
2. Given the player enters an incorrect printf command, the NPC will keep asking the player to re-enter the answer until the player enters the right answer.
3. Given a teaching module for printf, when the player interacts with the tutorial, then it must explain and demonstrate the usage of printf with char, string, int, and double.

4. Given the player uses the correct printf syntax for naming their character, when the name is entered, then it must update across the game and save to the database.
5. Given the game connects the frontend and backend, when the player finishes a naming and testing task, then the results must sync with the backend and persist in the database.

User Story #12

As a user, I want to see a figure on the map to see which level I'm currently on.

#	Description	Estimated Time	Owner
1	Find graph material	1hr	Xinyi Guan
2	Add map graph	1hr	Xinyi Guan
3	Add according level button(load from database) and say "You are here"	1hr	Xinyi Guan
4	Add several other level buttons	1hr	Xinyi Guan
5	Make level buttons distinguishable (completed, unlocked, and locked levels)	2hr	Xinyi Guan
6	Design and find map graph	1hr	Xinyi Guan
7	Testing	1hr	Xinyi Guan

Acceptance Criteria

1. Given the map graph material is loaded, when I open the map screen, then the map graph must display the current level visually.
2. Given the player's current level data is loaded from the database, when the map screen is opened, then the current level must be highlighted with a message stating, "You are here."
3. Given the map includes multiple levels, when I navigate through the map, then each level must display as a button.
4. Given other level buttons are added, when I view the map, then it must visually distinguish between completed, unlocked, and locked levels.

User Story #13

As a user, I would like to create multiple save files under one account, so I can have different game progressions.

#	Description	Estimated Time	Owner
1	Design "save file" database	2hrs	Xinyi Wu
2	Back-end logic for "save" all information	3hr	Xiuwen Fang
3	UI of save list page	2hr	Xiuwen Fang
4	Add popup window to enter name of save file	1hr	Xiuwen Fang
5	Connect front end and back end	1hr	Xiuwen Fang
6	Testing	1hr	Xiuwen Fang

Acceptance Criteria

1. Given that the user is on the main page after login, when the user clicks the "New Game" button, then a popup window should appear prompting the user to enter a name for the save file.
2. Given that the user has entered a save file name, when the backend successfully processes the request and stores the new save file in the database, then the new save file should be displayed in the save list.
3. Given that the user has already created one or more save files, when the user creates additional save files, then all save files should be listed on the save list page.
4. Given that the user has created save files, when the user logs out and logs back in, then the save files should still be displayed on the save list page.

User Story #14

As a user, I would like to load a specific save file from the list of my saves.

#	Description	Estimated Time	Owner
1	Logic for loading "save file" on the backend	3hr	Xiuwen Fang
2	connect front end and back end	1hr	Xiuwen Fang
3	UI loading save file data	2hr	Xiuwen Fang
4	Add Confirmation Popup	1hr	Xiuwen Fang
5	Test that different archives load correctly	2hr	Xiuwen Fang

Acceptance Criteria

1. Given that the user is on the save list page and has available saves, when the user clicks on "loading", then a loadable archive will appear and can be clicked on.
2. Given that the user has no save files in their account, when the user opens the save list page, then the system should display a message stating that no save files exist.
3. Given that the user has selected a save file, when the user clicks the "Load" button, then the system should display a confirmation popup asking if the user wants to load the file.
4. Given that the user has multiple save files, when the user selects a specific save file and clicks "Load," then the game should load the correct save data corresponding to the selected file.

User Story #15

As a user, I would like to delete save files that I no longer need.

#	Description	Estimated Time	Owner
1	Adding a Delete Button to the UI	0.5hr	Xiuwen Fang
2	Backend logic for deleting archived data	3hr	Xiuwen Fang
3	Connect front end and back end	1hr	Xiuwen Fang
4	Handling UI Refresh	2hr	Xiuwen Fang
5	Add Delete Confirmation Popup window	1hr	Xiuwen Fang
6	Testing to ensure that deleting an archive does not affect other archives	1hr	Xiuwen Fang

Acceptance Criteria

1. Given that the user is on the main page and the save list is displayed, when the user views the save list, then each save file in the list should have a corresponding "Delete" button.
2. Given that the user is on the main page and the save list contains one or more save files, when the user clicks the "Delete" button next to a save file, then a confirmation popup window should appear.
3. Given that the confirmation popup window is displayed, when the user clicks the "Confirm" button, then the system should send a request to the backend to delete the selected save file, And the popup window should close after the request is initiated.

4. Given that the frontend receives a success response from the backend, when the response indicates the deletion was successful, then the deleted save file should no longer appear in the list.

Total Working Hours for Sprint1

Member Name	Estimated Time
Xiuwen “Charlie” Fang	40 hrs
Xinyi Guan	42 hrs
Yuchuan Huang	45 hrs
Xinyi Wu	42 hrs
Total	169 hrs

Remaining Backlog:

Functional Requirements

- Welcome and Main Menu:

- ~~1. As a user, I would like to see a loading animation when the game starts, so that the game feels engaging and polished.~~
- ~~2. As a user, I want the option to skip the loading animation and directly access the main menu, so I can save time when re-entering the game.~~
- ~~3. As a user, I would like to see a main page after loading the welcome animation, including the function buttons I need(log-in, settings, etc.).~~
- ~~4. As a user, I would like to create my own account.~~
- ~~5. As a user, I would like to log in and log out of my own account.~~
6. As a user, I want to access settings to adjust volume and sound effects.
7. As a user, I would like to view statistical data(percentage) on game progress, such as card collections and features collection progress.

- Save Management:

- ~~8. As a user, I would like to create multiple save files under one account, so I can have different game progressions.~~
- ~~9. As a user, I would like to load a specific save file from the list of my saves.~~
- ~~10. As a user, I would like to delete save files that I no longer need.~~

- Achievements

11. As a user, I would like to unlock achievements with dates of receiving for significant milestones in the game, so I feel a sense of progression and accomplishment.
- ~~12. As a user, I want an on-screen notification when I unlock an achievement, so I feel immediate satisfaction and recognition.~~
13. As a user, I would like to earn rewards (e.g., in-game currency, items, or cosmetics) for completing achievements, so I feel incentivized to pursue them.
- ~~14. As a user, I want a dedicated page in the game to display all the achievements I have unlocked, so I can revisit and showcase my accomplishments.~~

- Game Background and Core Flow

15. As a user, I would like the option to pause the game and return to the main menu, without saving the progress.
16. As a user, I want to see an option to restart the current level from the pause menu, so I can quickly try again if I make a mistake.
17. As a user, I want to see a detailed breakdown of rewards (e.g., experience points, loot) on the victory settlement screen, so I know what I've earned.
18. As a user, I would like the failure screen to provide suggestions or tips for improving in the next attempt, so I feel motivated to try again.
19. As a user, I want the victory and failure screens to offer quick options to replay the level or return to the map screen, so I can easily decide my next step.

20. As a user, I would like a narrative story to provide background for the game world and context for my actions.

- User Page

21. As a user, I would like to view my profile information in the user page.

22. As a user, I want the ability to modify my personal details (e.g., username, email, and password) from the user page, so I can keep my account up to date.

23. As a user, I want to upload and update a custom profile picture, so I can personalize my account.

24. As a user, I want a quick link to contact support or provide feedback directly from my user page, so I can resolve issues or share suggestions conveniently.

- Card Game Mechanics

25. As a user, I would like to preview card attributes to understand their effects before using them.

26. As a user, I would like to select the functional cards I need before entering battle rounds to customize my strategy.

27. As a user, I would like to filter and sort cards by attributes (e.g., attack, defense, or special effects) when selecting my functional cards before a battle, so I can quickly find the cards I need.

28. As a user, I would like to save multiple custom decks for different battle scenarios, so I can switch strategies easily.

~~29. As a user, I want each card to have a specific amount of cost, so I need to carefully manage my resources during each turn.~~

~~30. As a player, I want to see my health, enemy health, and card details during combat so that I can make informed decisions.~~

~~31. As a user, I want a basic deck of cards so that I can draw and play cards in combat.~~

~~32. As a player, I want to engage in a simple turn-based combat system so I have time to plan my strategy and respond to the boss's moves.~~

33. As a user, I want the boss to have unique abilities or cards that challenge my strategy, so the battles feel varied and exciting.

- Map and Exploration

34. As a user, I want to unlock new map areas by either completing specific levels or learning certain skills, so I can explore the game world in a way that aligns with my progress and choices.

~~35. As a user, I want to see a figure on the map to see which level I'm currently on.~~

36. As a user, I want the map to restrict access to incomplete levels while allowing me to freely revisit previously completed levels, so I can replay them for practice or better rewards.

37. As a user, I want to include optional levels in the map which focus on specific skills or learning topics, so I can choose to strengthen areas that interest me.

38. As a user, I want to zoom in and out of the map with ease, so I can navigate between detailed sections and the overall layout effortlessly.
39. As a user, I want to unlock map-themed achievements, so I feel motivated to explore thoroughly.
- Reward System
40. As a user, I would like to earn rewards for completing levels and challenges, which can be used to exchange for blank functional cards or other upgrades.
41. As a user, I would like to receive different types of rewards (e.g., blank cards, rare items, in-game currency, or exclusive abilities), so my gameplay feels more varied and engaging.
42. As a user, I would like bonus rewards for completing levels with certain conditions (e.g., within a time limit or without losing health), so I can aim for more ambitious goals.
43. As a user, I would like the rewards for challenges to scale with difficulty, so I feel incentivized to take on harder challenges.
- Learning Features
 - Learning printf:
- ~~44. As a user, I would like to name my character while learning the printf function, so I can understand how to format strings and personalize my game experience.~~
- ~~45. As a user, I want the game to have NPC interactions to teach me the basics of the printf function, so I can use them effectively.~~
- ~~46. As a user, I would like mini-games to practice printf with increasingly complex printf challenges, so I can gradually improve my skills.~~
- ~~47. As a user, I want to earn my first achievement, "Person You Know Who," by demonstrating mastery of the printf function and shown on one achievement page, so I feel motivated to continue learning.~~
- Learning Basic Math:
48. As a user, I want a blacksmith NPC to teach me basic math operations (addition, subtraction, multiplication, division) through interactive examples, so I can learn while engaging with the story.
49. As a user, I would like to complete an independent quest helping a NPC manage their finances, where I apply math skills to solve real-world-like problems.
50. As a user, I want to unlock the in-game shop and earn the achievement "Live for your own" after completing a series of math-based challenges, so I feel a sense of accomplishment.
51. As a user, I want the math challenges to scale in difficulty, starting with simple calculations and progressing to multi-step problems, so I can gradually improve my skills.
- Learning if-else and for/while Loops:
52. As a user, I want to learn conditional statements and loops through a maze game where NPCs provide guided examples, so I can practice these concepts in a visual and interactive way.

53. As a user, I want to earn rewards for solving maze challenges using if-else statements and loops, so I feel motivated to apply these concepts effectively.
54. As a user, I want the looping challenges to scale in difficulty, like with different maze maps, so I can gradually improve my skills.

- Learning Pointers:

55. As a user, I would like a quest involving keys, chests, and addresses to learn the basics of pointers and callback functions, so I can understand how to reference and use memory locations in a practical context.
56. As a user, I want an NPC to explain pointers with visual examples (e.g., diagrams showing memory addresses and values), so I can better grasp the concept.
57. As a user, I want to unlock pointer-related enchantments (e.g., dynamic item enhancements) after completing the quest, so I feel rewarded for mastering a complex concept.
58. As a user, I want to receive a specific achievement or rewards after learning pointers.

- Shop and Blacksmith Features

59. As a user, I want to spend earned money in a shop to purchase blank functional cards, so I can customize my deck and strategy.
60. As a user, I want the blacksmith to enchant functional cards with achieved functions, so my cards can gain new abilities and become more versatile.
61. As a user, I want access to magic cards that provide permanent stamina boosts or extra coin rewards, so I can gain long-term advantages in the game.
62. As a user, I would like to upgrade my functional cards for better performance, so they are more effective during battles.
63. As a user, I want the shop to offer limited-time items (e.g., rare cards, consumables) that rotate daily or weekly, so I feel incentivized to check back regularly.
64. As a user, I would like to trade in unused cards or items for coins or shop credits, so I can make better use of my resources.